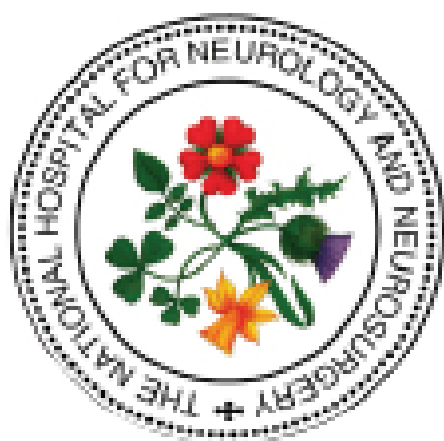




THIOPENTAL for REFRACTORY RAISED ICP in ACUTE BRAIN INJURY

INDICATIONS FOR USE

Thiopental is a rapidly acting barbiturate used mainly for induction of anaesthesia

It can be used in patients with acute brain injury to control intractable intracranial hypertension (by decreasing the metabolic demands of the brain) when other conventional medical and simple surgical methods to lower ICP are ineffective. It may be used as a last resort before carrying out decompressive craniectomy.

Thiopental is only commenced when all other conventional medical measures to control ICP have failed.

STABILITY AND COMPATIBILITY

- Reconstituted solutions of Thiopental should be discarded after 24 hours
- Thiopental should not be given with any other drugs
- Before commencing ensure patient adequately hydrated as hypovolaemia may precipitate hypotension

COMPLICATIONS OF PROLONGED THERAPY

Prolonged clinical effect especially in patients with impaired renal / hepatic function

Hypotension – implications for CPP therefore reduce other sedative drugs

Loss of pupillary reactivity – pupils can become fixed

For brain-stem testing, levels must be obtained (*thiopental level must be <5mg/l – but a level of **zero** might be required if patient has acute kidney injury*)

HIGH DOSE THIOPENTAL in REFRACTORY INTRACRANIAL HYPERTENSION

ONLY USED WHEN ALL OTHER MEASURES HAVE FAILED TO CONTROL ICP

STEP 1

- Thiopental is administered as a **25mg/ml** (2.5%) solution
- Reconstitute each 500mg vial with 20ml of water for injection
- For infusions - reconstitute 3 x 500mg vials of Thiopental with 60ml of water for injection
- Preferably given via a central line

STEP 2

LOADING DOSE

- Give **250mg** IV bolus over **5 minutes**
- Repeat until a reduction in ICP is observed – up to a **maximum of 3 bolus doses**
- When a reduction is seen commence high dose infusion (see below)
- *Bolus acts within one arm-brain circulation time (i.e. 1 minute), effect lasts for 5-15 minutes and is cumulative after repeated administration*

If no reduction in ICP after 3 bolus doses infusion should NOT be commenced

STEP 3

INFUSION

- Commence infusion at **3-5mg/kg/h** (*Higher doses must be discussed with NCC consultant prior to increasing infusion*)
- Infusion rate should be titrated to achieve **ICP <22mmHg**
- If infusion is effective other sedation should be **reduced** to maintain CPP and limit noradrenaline requirements
- Infusion should continue for **maximum 48 hours** without further discussion with NCCU Consultant